

Problem F. Sereja and D

Time Limit 1000 ms

Code Length Limit 50000 B

OS Linux

Read problems statements in [Mandarin Chinese](#) and [Russian](#).

Sereja has an array consisting of n integers $a_1 \leq a_2 \leq \dots \leq a_n$. Based on this array, he has to answer m queries represented by pairs integers t and d .

The answer for a query is the smallest integer i for which there exist some k ($i \leq k$) such that $a_i + d \geq a_{i+1}$, $a_{i+1} + d \geq a_{i+2}$, $\dots$, $a_{k-1} + d \geq a_k$, $a_k \leq t$ and $a_{k+1} > t$ (if it exists).

Help Sereja find the answer for each query.

Input

- First line of input contains a single integer n
- Next line contain n space separated integers $a_1, a_2, \dots, a_n$.
- Next line contains a single integer m .
- Next m lines contain pairs of integers representing the queries.

Output

- For each query, output the answer on a separate line.

Constraints

- $1 \leq n, m \leq 10^5$.
- $1 \leq a_i \leq 10^6$
- $a_1 \leq t \leq 10^6$
- $0 \leq d \leq 10^6$

Subtasks

Subtask 1 (30 points): $1 \leq n, m \leq 10^4$

Subtask 2 (70 points): No additional constraints.

Example Input

```

1 | 5
2 | 1 2 3 10 50
3 | 6
4 | 1 1
5 | 5 3
6 | 11 7
7 | 100000 1
8 | 1000000 1000000
9 | 11 6

```

Example Output

```

1 | 1
2 | 1
3 | 1
4 | 5
5 | 1
6 | 4

```